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Description - This procedure will be used by all Signa Systems that have access to the RF Calculation Conversion Software Tools (QRFCALC and RF CALC).

1- OVERVIEW

This procedure describes the use of the RF Calculation Conversion Software Tools (QRFCALC and RF CALC). An example of each tool being used is provided.
Additionally, this procedure provides an example of the RF Power Measurement Kit being used with the RF Calculation Conversion Software Tools (QRFCALC and RF CALC).

Note

All 1.5T systems using the RF Power Measurement Kit must be calibrated (the oscilloscope adjusted to 8 Divisions to remove frequency roll-off error/bandwidth limitations) with the 1.5T Scope Calibrator. If an oscilloscope does not allow the user to adjust display (via the Volts/Variable knob on the oscilloscope) to 8 Divisions when using the 1.5T Scope Calibrator, a separate process has been provided.

2- TOOLS AND INSTRUMENTS REQUIRED

Refer to Table 1 and Illustration L2813A.

TABLE 1
TOOLS AND INSTRUMENTS REQUIRED

Description	Part Number	Qty.
RF Power Measurement Kit	46-317724G1 or G2	1
Magnetic Shield (Not Shown)	46-317725G10	1

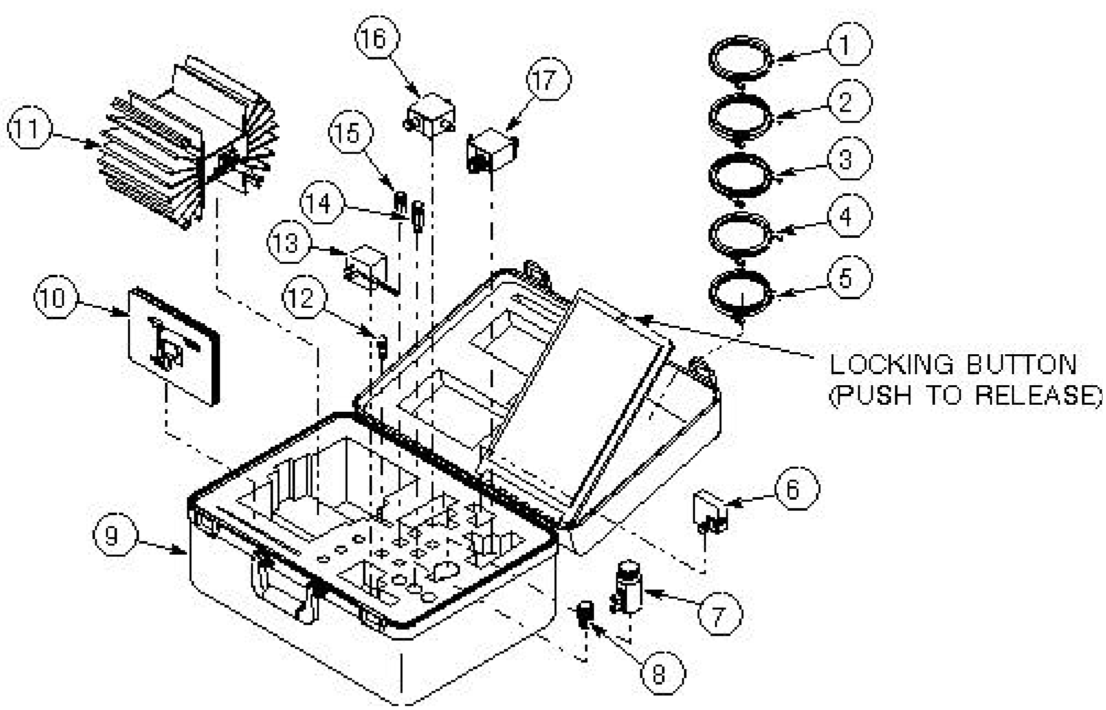

ILLUSTRATION L2813A
RF POWER MEASUREMENT KIT

2- OSCILLOSCOPES WITHOUT VOLTS/VARIABLE ADJUSTMENT CAPABILITY

1. Procure the RF Power Measurement Kit.
2. Refer to the ‘CAL’ Card provided in the RF Power Measurement Kit for hardware set-up ONLY.

Note

All 1.5T systems using the RF Power Measurement Kit must be calibrated (the oscilloscope adjusted to 8 Divisions to remove frequency roll-off error/bandwidth limitations) with the 1.5T Scope Calibrator.

If an oscilloscope does not allow the user to adjust display (via the Volts/Variable knob on the oscilloscope) to 8 Divisions when using the 1.5T Scope Calibrator, a separate process has been provided.

CAUTION

Possible measurement error. Measuring RF power by using an oscilloscope without the use of the magnetic shield may cause serious measurement errors. The magnetic shield must be used in RF power measurements.

1. Install the oscilloscope (Tektronix 468 or 2465B) into the magnetic shield (46-317725G10). See Illustration L2814A.

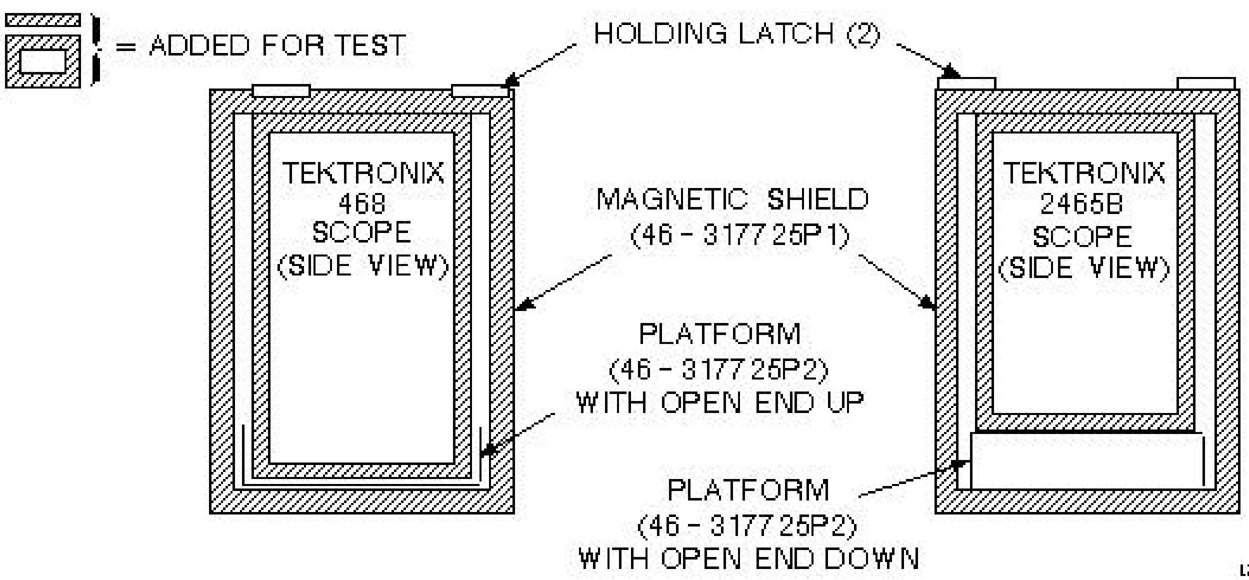

ILLUSTRATION L2814A
OSCILLOSCOPE MAGNETIC SHIELD CONFIGURATION

Note

Tektronics 2230 and 2232 are not the MR recommended oscilloscopes, but can be used with the magnetic shield by removing the scope handles using a torx screwdriver.

Note

The scope/magnetic shield unit may be lifted by the scope handle for purposes of moving the unit.

Description - A set of reference cards is provided to illustrate hardware setup using the RF Power Measurement Kit. Two software tools, RFCALC and QRFCALC, are available to convert between Watts, dBm, and peak amplitude (VPP referred to as Scope Reading on 0.5T and 1.0T Kit Cards) (Div referred to as Scope Reading on 1.5T Kit Cards).

CAUTION

All 1.5T systems using the RF Power Measurement Kit must be calibrate (the oscilloscope to 8 Divisions to remove frequency roll-off error/bandwidth limitations) with the 1.5T Scope Calibrator. Refer to the 'CAL' Card provided in the RF Power Measurement Kit.

5- REFERENCE CARD USE

The RF Power Measurement Kit contains color coded reference cards which depict all power measurement schemes for 5.x systems. Refer to Table 3 for 1.5T and Table 6 for 1.0T. The kits come calibrated with all attenuation loss values established and printed on each card. See Illustration L2817A in Section 2. Power level readings taken from the scope, along with attenuation loss value, can be entered into a software program (QRF CALC) to determine the actual RF source power level.

TABLE 3HOTWORDSTYLE=BOOKDEFAULT;
QUICK REFERENCE CARDS PROCEDURAL USAGE (1.5T)

Card #	Reference Card Title	Procedure Used With	Card 46#
72 & 63	BODY OUTOUT 72dBm (16KW) & HEAD OUTPUT 63dBm (2KW)	procedure for APB Calibration	46-317724P17 & 46-317724P18
CAL	SCOPE CALIBRATION +4dBm 46-317724P15	procedure for APBCalibration procedure for Power Monitor Check	
52	ERBTEC SOLID STATE OUTPUT (HEAD MODE) 52dBm (160W)	TROUBLESHOOTING PROCEDURE TBD	46-317724P20
49	ERBTEC SOLID STATE OUTPUT (BODY MODE) 49dBm (80W)	TROUBLESHOOTING PROCEDURE TBD	46-317724P21
20	ERBTEC RF TEST OUTPUT 20dBm (0.1W)	TROUBLESHOOTING PROCEDURE TBD	46-317724P23
10	EXCITER RF OUTPUT 10dBm (0.01W)	procedure for APB Calibration	46-317724P24
KIT	KIT CONTENTS & PLACEMENT GUIDE	N/A	46-317724P16

TABLE 6HOTWORDSTYLE=BOOKDEFAULT;
QUICK REFERENCE CARDS PROCEDURAL USAGE (1.0T)

Card #	Reference Card Title	Procedure Used With	Card 46#
68.8 & 58.8	BODY OUTOUT 68.8dBm (7.58KW) & HEAD OUTPUT 58.8dBm (750W)	procedure for APBCalibration	46-317724P32 & 46-317724P33
52	ERBTEC SOLID STATE OUTPUT (HEAD MODE) 52dBm (160W)	TROUBLESHOOTING PROCEDURE TBD	46-317724P35
55	ERBTEC SOLID STATE OUTPUT (BODY MODE) 55dBm (320W)	TROUBLESHOOTING PROCEDURE TBD	46-317724P34
20	ERBTEC RF TEST OUTPUT 20dBm (0.1W)	TROUBLESHOOTING PROCEDURE TBD	46-317724P36
10	EXCITER RF OUTPUT 10dBm (0.01W)	procedure for APB Calibration	46-317724P37
KIT	KIT CONTENTS & PLACEMENT GUIDE	N/A	46-317724P16

CAUTION

Ensure that the output of the attenuator is terminated with the 50- ohm terminator chained to the attenuator to create a dummy load as illustrated on the reference cards. See Illustration L2818A. If the attenuator is not terminated, RF signal can be transmitted into the surrounding area and erroneous measurement errors can occur.

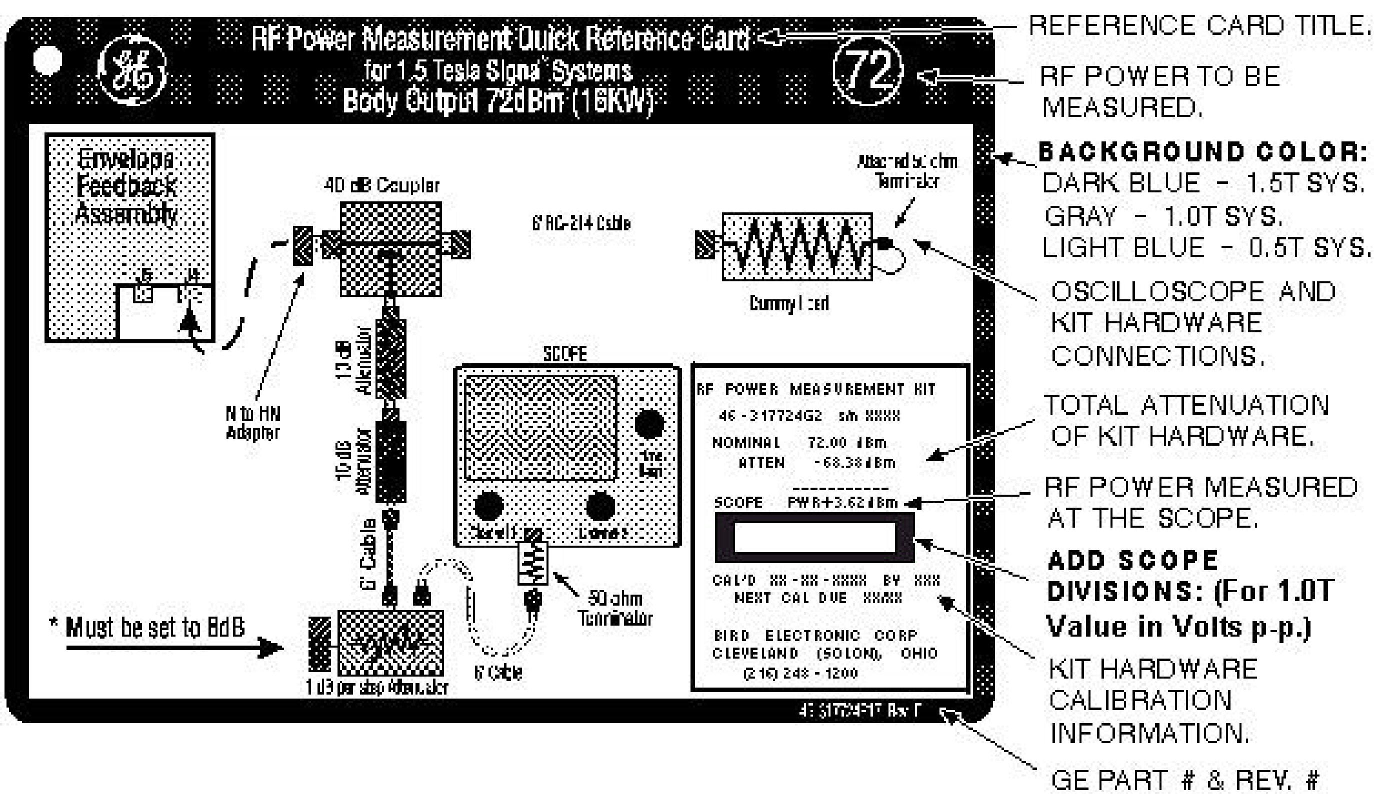

ILLUSTRATION L2818A
RF POWER MEASUREMENT QUICK REFERENCE CARD

Note

The total attenuation listed on the reference cards does not include the scope or dummy load and its interconnecting cable.

Description - The oscilloscope calibrator tool accurately calibrates all oscilloscopes to allow measurement of RF power at 63.86 MHz. A set of reference cards is used to illustrate setup. Two software tools, RFCALC and QRFCALC, are available to convert between watts, dBm, and peak amplitude.

When making any RF measurements, all cables must be 50 ohms and terminated into 50 ohms. In RF signal measurements, power is often rendered in dBm (decibels referenced to 1 milliwatt) instead of in watts. With the RF system, 16kW equals +72 dBm, and 2kW equals +63 dBm. See Illustration L2812A.

$$\text{Power (W)} = (V_{p-p} / (2\sqrt{2}))^2 / 50 \quad \text{where } V_{p-p} = 2V_{peak}$$

Ex: $$2.2 = (30 / (2\sqrt{2}))^2 / 50$$

$$\text{Power (dBm)} = 20 \log (V_{p-p}) + 4$$

Ex: $$33.5 = 20 \log (30) + 4$$

$$\text{Power (dBm)} = 10 \log (\text{Power (watts)} / 0.001 \text{ (W)})$$

Ex: $$72.0 = 10 \log (16000 / 0.001 \text{ (W)})$$

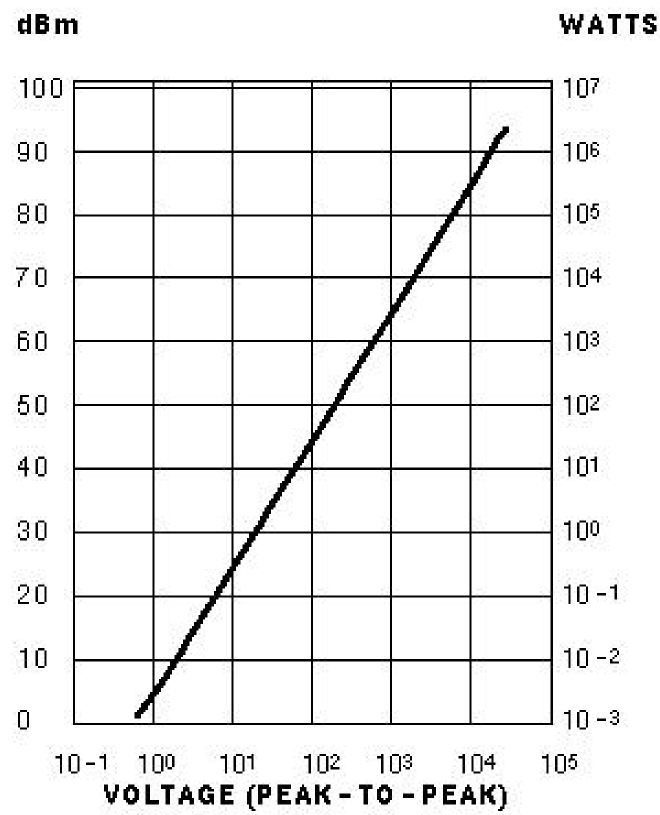

ILLUSTRATION L2812A
RF POWER MEASUREMENT

The decibel (dB) is a relative unit. When using decibels to specify an absolute value of power level, you must qualify the decibel value by a reference level. For example, in discussions of RF amplifier, a reference level of 0 dB can be considered 16kW. As the power is cut in half, the dB value is subtracted by 3 dB. (16kW=0dB, 8kW= -3dB, 4kW= -6dB, etc.)

2. **For 1.5T only:** Connect the oscilloscope calibrator to the oscilloscope as shown in Illustration L2815A.

Note

The Oscilloscope does not need to be specially calibrated for 1.0T RF power measurements. Read any RF signal directly off the scope screen in volts/div.

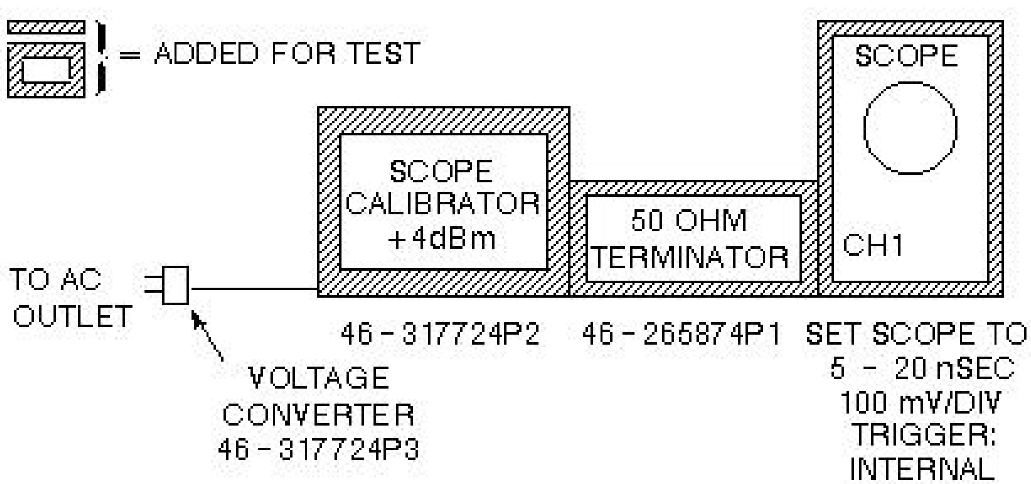

ILLUSTRATION L2815A

OSCILLOSCOPE CALIBRATOR CONFIGURATION

3. **For 1.5T only:** Adjust the VERTICAL VOLTS/DIV control until the amplitude of the waveform is slightly greater than 8 divisions. See Illustration L2816A.

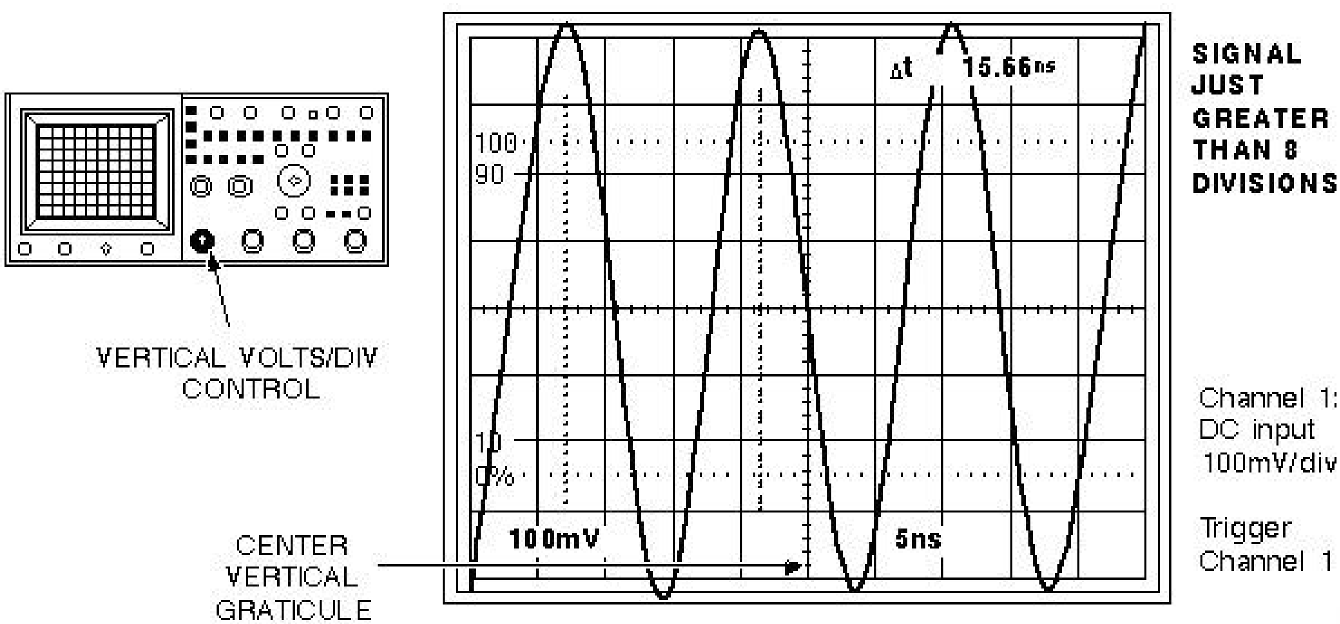

ILLUSTRATION L2816A

VERTICAL SCALE CONTROL WAVEFORM ADJUSTMENT

Note

Even with the magnetic shield, the oscilloscope waveform may be slightly distorted. Consistently use the area near the center vertical graticule to measure the RF power signals. (See Illustration L2816A.)

4. **For 1.5T only:** Adjust the vertical volts/div VARIABLE control to achieve exactly 8 divisions. See Illustration L2817A.

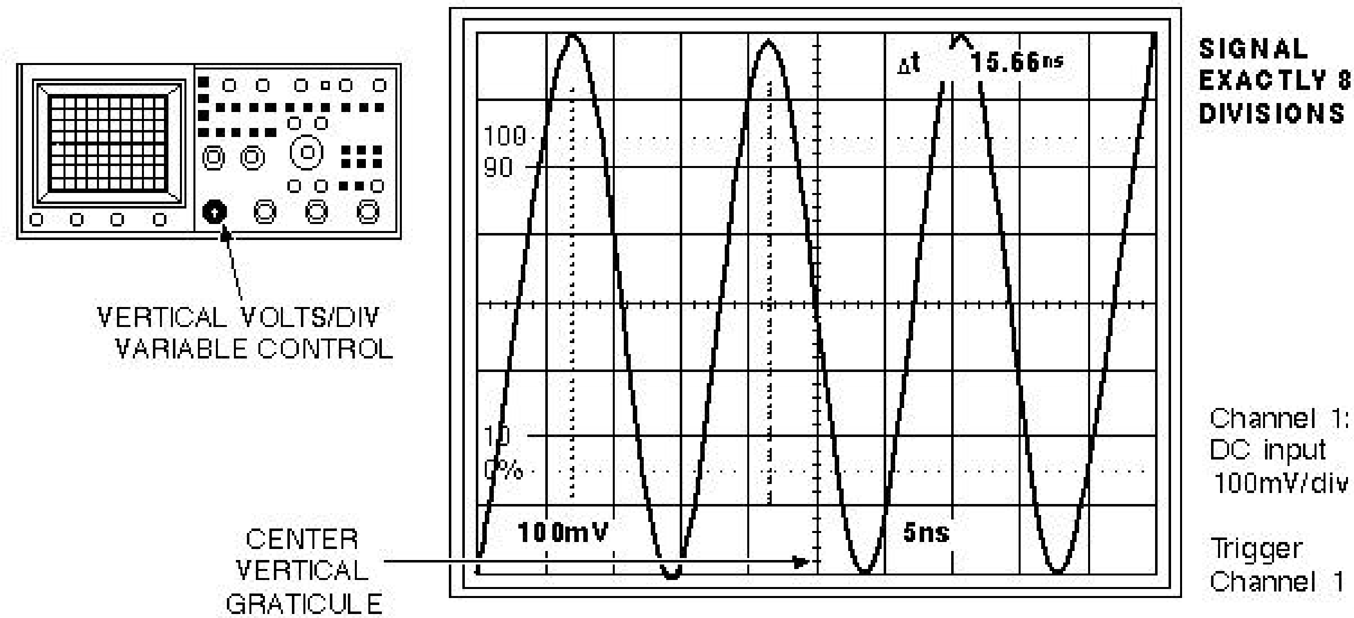

ILLUSTRATION L2817A
VARIABLE CONTROL WAVEFORM ADJUSTMENT

Note

Even with the magnetic shield, the oscilloscope waveform may be slightly distorted. Consistently use the area near the center vertical graticule to measure the RF power signals. (See Illustration L2817A.)

CAUTION

This procedure has calibrated the oscilloscope for a specific Volts/Div scale. Do not move the VERTICAL VOLTS/DIV or VARIABLE controls after calibration, or the RF power measurements will be in error.

5. **For 1.5T only:** Disconnect ac voltage converter from power and remove the +4dBm calibrator. Leave the 50 ohm terminator attached to the oscilloscope.

Note

It may be necessary to change the oscilloscope timebase to 2 msec or slower to view the entire RF signal during calibration and troubleshooting procedures. Use the procedure's recommended timebase where noted.

3- QUICK RF CALCULATION (QRF CALC) CONVERSION SOFTWARE TOOL

1. From the MR Tools desktop, select **[Cals/checks]** and **[RF Calc]**.

Note

It may be necessary to change the oscilloscope timebase to 2 msec or slower to view the entire RF signal during calibration and troubleshooting procedures. Use the procedure's recommended timebase where noted.

See Table 7

TABLE 7
QUICK RF CALCULATION (QRF CALC) CONVERSION SOFTWARE TOOL

Output/Prompts	Input/Comments

`(Q)RF Calculator`
`(R)F Calculator`
`Enter selection: (Q,R) [Q]:...........`
	<Enter> *To select QRFCALC. (Do not select Q which indicates QUIT.)*

`Quick RF Power Calculator`
`(A)mplitude from power`
`(P)ower from amplitude`
`(Q)uit`
`Enter selection: (A,P) [A]:...........`
	A <Enter> *To select Amplitude.*

`Convert from Power to Scope Amplitude`	

Note

For each reference card SCOPE PWR value, record the above equivalent scope amplitude (in divisions) on each reference card, just below the SCOPE PWR value. In the future, the scope power in divisions will be calculated and printed on all reference cards by the vendor. See Illustration L2817A in Section 2.

Note

Perform POWER FROM AMPLITUDE to calculate the source power, given the measured scope divisions of the RF signal and the attenuation factor of the hardware used. (Refer to Section 5- Reference Card Use). If the source power is already known, continue with the next section.

See Table 8

TABLE 8
QUICK RF CALCULATION (QRFCALC) CONVERSION SOFTWARE TOOL

Output/Prompts	Input/Comments
Quick RF Power Calculator (A)mplitude from power (P)ower from amplitude (Q)uit	
Enter selection: (A,P) [A]:..........	**P <Enter>** *To select Power from amplitude.*
Convert from Scope Amplitude to Power Oscilloscope calibrated (with tool) to ...: 4.00 (dBm) with a reference amplitude of............: 8.00 (divisions)	

See Table 5, QRF CALC Algorithms:

TABLE 5
TABLE 5: QRF CALC ALGORITHMS

QRF CALC ALGORITHMS:

Amplitude (divisions) from Power (dBm)

$$\text{Volts (desired [in divisions])} = 10^{\frac{\text{dbm(desired)} - 4.00}{20.0}} \qquad \text{Ex:} \quad 7.1 = 10^{\frac{3.0 - 4.00}{20.0}} \times 8.0$$

Power (dBm) from Amplitude (divisons)

$$\text{Power (dBm)} = 20 \log (V_{p\text{-}p} \text{ [in divisions]}) + 4 \qquad \text{Ex:} \quad 33.54 = 20 \log (30) + 4$$

Power (dBm) including Attenuation

$$\text{Source Power (dBm)} = \text{Total Attenuation (dB)} + \text{Scope Power Reading (dBm)} \qquad \text{Ex:} \quad 72.25 = 68.5 + 3.75$$

4- RF CALCULATION (RFCALC) CONVERSION SOFTWARE TOOL

1. From the MR Tools desktop, select **[Cals/checks]** and **[RF Calc]**.

Note

Perform WATTS TO DBM or DBM TO WATTS to convert a RF power signal from watts to dBm or dBm to watts. For relative voltage to relative power conversion, continue with the next page.

See Table 9

TABLE 9
RF CALCULATION (RFCALC) CONVERSION SOFTWARE TOOL

Output/Prompts	Input/Comments
(Q)RF Calculator	
(R)F Calculator	
Enter selection: (Q,R) [Q]:.........	**R<Enter>** *To select RF CALC.*
RF Power Calculator	
1)Watts to dBm	
2)dBm to Watts	
3)Scope relative voltage from relative power (Watts)	
4)Scope relative voltage from relative power (dBm)	
5)Scope relative power from relative voltage (Watts)	
6)Scope relative power from relative voltage (dBm)	
7)Quit	

Note

Perform SCOPE RELATIVE VOLTAGE FROM RELATIVE POWER (WATTS or DBM) to convert a RF power signal from voltage to watts or dBm based on the ratio of a known (reference) voltage to power equivalent. For example, if RF power is being measured with a 400 MHz scope, the following conversion can be used to determine the correct value for a 100 MHz scope. For relative power to relative voltage conversion, continue with the next page.

See Table 10

TABLE 10
RF CALCULATION (RFCALC) CONVERSION SOFTWARE TOOL

Output/Prompts	Input/Comments
RF Power Calculator 1) Watts to dBm 2) dBm to Watts 3) Scope relative voltage from relative power (Watts) 4) Scope relative voltage from relative power (dBm) 5) Scope relative power from relative voltage (Watts) 6) Scope relative power from relative voltage (dBm) 7) Quit	
Enter selection: (1..7) [1]:...........	**3<Enter>** To select Scope relative voltage from relative power (Watts).
****** NOTE: Constant attenuation is assumed! ***** Enter reference power in Watts: (0.0..20000.00) [16000.00]..................	**xxx.xx<Enter>** Enter reference power in Watts from reference card.
Enter reference Volts peak: (0.0..100.00) [10]..........................	**xx.xx<Enter>** Enter reference voltage peak("scope reading" from ref card).
Enter desired power in Watts: (0.0..20000.00) [10.00].....................	**xxx.xx<Enter>** *Enter desired power in Watts.*
Peak voltage for desired power is xx.xxx Volts	<---*Peak voltage desired.*
Press ENTER to continue [ ]:................	**<Enter>**

Note

Perform SCOPE RELATIVE POWER (WATTS or DBM) FROM RELATIVE VOLTAGE to convert a RF power signal from watts or dBm to voltage based on the ratio of a known (reference) power to voltage equivalent. For example, if RF power is being measured with a 400 MHz scope, the following conversion can be used to determine the correct value for a 100 MHz scope.

See Table 11.

TABLE 11
RF CALCULATION (RFCALC) CONVERSION SOFTWARE TOOL

Output/Prompts	Input/Comments
RF Power Calculator 1) Watts to dBm 2) dBm to Watts 3) Scope relative voltage from relative power (Watts) 4) Scope relative voltage from relative power (dBm) 5) Scope relative power from relative voltage (Watts) 6) Scope relative power from relative voltage (dBm) 7) Quit	
Enter selection: (1..7) [1]:........	**5<Enter>** *To select Scope relative power from relative voltage (Watts).*
****** NOTE: Constant attenuation is assumed! ******	
Enter reference power in Watts: (0.0..20000.00) [18000.00]..................	**xxx.xx<Enter>** *Enter reference power in Watts from reference card..*
Enter reference Volts peak: (0.0..100.00) [19.63].......................	**xx.xx<Enter>** *Enter reference voltage peak ("scope reading" from ref card).*
Enter measured volts peak: (0.0..100.00) [10.00].......................	**xxx.xx<Enter>** *Enter measured volts peak.*
Measured power is xx.xxx Watts	*<----Power desired in watts.*
Press ENTER to continue [ ]:................	**<Enter>**
RF Power Calculator 1) Watts to dBm	

2) dBm to Watts 3) Scope relative voltage from relative power (Watts) 4) Scope relative voltage from relative power (dBm) 5) Scope relative power from relative voltage (Watts) 6) Scope relative power from relative voltage (dBm) 7) Quit Enter selection: (1..7) [1]:.........	**6 <Enter>** *To select Scope relative power from relative voltage (dBm).*
****** NOTE: Constant attenuation is assumed! ****** Enter reference power in dBm: (0.0..1000.00) [72.60]......................	**xxx.xx <Enter>** *Enter reference power in dBm. This is the circled value in the upper right hand corner or the reference card.*
Enter reference Volts peak:(0.0..100.00) [19.63].......................	**xx.xx <Enter>** *Enter reference voltage peak.*
Enter measured volts peak: (0.0..100.00) [10.00].......................	**xxx.xx <Enter>** *Enter measured volts peak.*
Measured power is xx.xxx dBm	*<---Power desired in dBm.*
Press ENTER to continue [ ]:................	**<Enter>**
RF Power Calculator 1) Watts to dBm 2) dBm to Watts 3) Scope relative voltage from relative power (Watts) 4) Scope relative voltage from relative power (dBm) 5) Scope relative power from relative voltage (Watts) 6) Scope relative power from relative voltage (dBm) 7) Quit Enter selection: (1..7) [1]:.........	**7<Enter>** *To quit.*
To Exit Shell, type exit at the prompt.	

See Table 4.

TABLE 4
RF CALCULATION (RFCALC) CONVERSION SOFTWARE TOOL

RF CALC ALGORITHMS:

Watts to dBm

Power (dBm) = 10 log (Power [watts] x 1000) Ex: 72.04 = 10 log (16000 x 1000)

dBm to Watts

$$\text{Watts} = \frac{10^{\frac{\text{dBm}}{10.0}}}{1000} \qquad \text{Ex: } 15848 = \frac{10^{\frac{72}{10.0}}}{1000}$$

Scope relative voltage from relative power (Watts)

$$\text{Volts} = \sqrt{\frac{\textit{Watts (desired)}}{\textit{Watts (ref.)}}} \times \text{Volts (ref.)} \qquad \text{Ex: } 2262.4 = \sqrt{\frac{12800}{20000}} \times 2828$$

6- RESTORATION CHECK LIST

❑ - Disconnect all kit components used in measurement scheme and return to kit. (Refer to "KIT" card for placement.)

❑ Restore all system cables and components to original configuration.

CAUTION

Possible measurement errors. Scope measurements can be in error if Vertical Volts/Div Variable control is not reset to calibrated condition. Ensure that the Vertical Volts/Div Variable control is reset to calibrated condition when not measuring RF signal power.

❑ Adjust the vertical volts/div VARIABLE control clockwise until it clicks into the calibrated condition.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	July 22, 1998	M.Whitlow	Initial conversion from Toolbook to Word.